

1. IDENTIFICATION OF PRODUCT AND COMPANY

Product Name : 77001-00030 (77001-00001) Solvent

Use of the substance/preparation: Solvent for industrial ink jet printers (CIJ-printers)

Supplier : Paul Leibinger Inc
2702 Buell Drive, Suite B
East Troy, WI 53120, USA
Phone: +01-262-642-4030
www.leibinger-group.com

Emergency phone (24h service) : USA: ChemTel Inc: +1-800-255-3924
Outside USA: ChemTel Inc: +1-813-248-0585
Brazil: ChemTel Inc: +1-800-591-5042
Mexico: ChemTel Inc: +1-800-099-0731

2. HAZARDS IDENTIFICATION

***** EMERGENCY OVERVIEW ***:** Highly flammable liquid and vapor. Causes serious eye irritation.

GHS Classification

<u>Hazard classes and Hazard categories</u>	<u>Hazard Statements</u>
Flammable Liquid, category 2	H225
Eye Irritation, category 2	H319
STOT, single exposure, category 3, NE	H336

Labelling according to Regulation (EC) No 1272/2008 [CLP/GHS]



Signal word: Danger

GHS Hazard statements

H225 Highly flammable liquid and vapour.
H319 Causes serious eye irritation.
H336 May cause drowsiness or dizziness.

GHS Precautionary Statements

P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P240 Ground/bond container and receiving equipment.
P241 Use explosion-proof electrical/ventilating/lighting/equipment.
P242 Use only non-sparking tools.
P243 Take precautionary measures against static discharge.
P261 Avoid breathing dust/fume/gas/mist/vapors/spray.
P264 Wash thoroughly after handling.
P271 Use only outdoors or in a well-ventilated area.
P280 Wear protective gloves/protective clothing/eye protection/face protection.
P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.
P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P312 Call a POISON CENTER/doctor/physician if you feel unwell.
P337+P313 If eye irritation persists: Get medical advice/attention.
P370+P378 In case of fire: Use appropriate method to extinguish.
P403+P233 + P235 Store in a well-ventilated place. Keep cool. Keep container tightly closed.
P405 Store locked up.
P501 Dispose of contents/container in accordance with local/regional/national/international regulations.

3. COMPOSITION / INFORMATION ON INGREDIENTS

Hazardous ingredients

CAS No	Name	[% weight]	Classification according to Regulation GHS
78-93-3	2-butanone	80-100	Flam. Liq. 2, H225 Eye Irrit. 2, H319 / STOT SE 3, H336
67-64-1	2-propanone	2-10	Flam. Liq. 2, H225 Eye Irrit. 2, H319 / STOT SE 3, H336

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

4. FIRST AID MEASURES



FIRST AID - EYE CONTACT: Immediately flush eyes with water. Flush eyes with water for a minimum of 15 minutes, occasionally lifting and lowering upper lids. Get medical attention promptly. Remove contact lenses if worn.

FIRST AID - SKIN CONTACT: Immediately flush skin with plenty of water. Remove clothing. Get medical attention immediately. Wash clothing separately and clean shoes before reuse.

FIRST AID - INHALATION: Rescuers should put on appropriate protective gear. Remove from area of exposure. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Keep victim warm. Get immediate medical attention. To prevent aspiration, keep head below knees.

FIRST AID - INGESTION: Small amounts which accidentally enter mouth should be rinsed out until taste of it is gone. DO NOT induce vomiting. Never give anything by mouth to an unconscious person. Consult a physician.

5. FIRE-FIGHTING MEASURES

UNUSUAL FIRE AND EXPLOSION HAZARDS: Highly flammable liquid and vapor. Vapors/dust may form explosive mixture with air.

Vapors can travel to a source of ignition and flash back. Empty containers retain product residue (liquid and/or vapor) and can be dangerous. DO NOT pressurize, cut, weld, braze, solder, drill, grind, or expose such containers to heat, flame, sparks, static electricity, or other sources of ignition. Also, do not reuse container without commercial cleaning or reconditioning. Closed container may explode under extreme heat.

SPECIAL FIREFIGHTING PROCEDURES: As in any fire, wear self-contained breathing apparatus pressure-demand (MSHA/NIOSH approved or equivalent) and full protective gear. Evacuate all unnecessary personnel. Shut down motors, pumps, electrical service and eliminate all sources of ignition. Avoid use of solid water streams. Do not use water jet (frothing possible). Use water with caution. Material will float and may ignite on surface of water. Water may be ineffective in fighting the fire. Water spray to cool containers or protect personnel. Use with caution. Water runoff can cause environmental damage. Dike and collect water used to fight fire. Small fires: carbon dioxide or dry chemical. Large fire: alcohol-type aqueous film-forming foam or water spray.

EXTINGUISHING MEDIA: Carbon Dioxide, Dry Chemical, Foam, Water Fog

6. ACCIDENTAL RELEASE MEASURES

STEPS TO BE TAKEN IN CASE MATERIAL IS RELEASED OR SPILLED: Wear appropriate personal protective equipment. (See Exposure Controls / Personal Protection Section.) Eliminate all ignition sources. Evacuate unnecessary personnel. Prevent additional discharge of material if able to do so safely. May cause slipping hazard. Do not touch or walk through spilled material.

Avoid runoff into storm sewers and ditches which lead to waterways. Ventilate spill area. Stay upwind of spill. A vapor suppressing foam may be used to reduce vapors. Collect spilled materials for disposal. Use only non-combustible material for clean-up. Use clean, non-sparking tools to collect absorbed materials. Absorb spill with inert material (e.g. dry sand or earth), then place in a chemical waste container. Recover by pumping (use an explosion proof or hand pump).

7. HANDLING AND STORAGE



HANDLING: Use only in a well ventilated area. Avoid breathing vapor, fumes or mist. Avoid contact with eyes, skin, and clothing. Take precautionary measures against static discharge. When transferring, follow proper grounding procedures. Use spark-resistant tools. Do not load into compartments adjacent to heated cargo. Use explosion proof equipment. Always open containers slowly to allow any excess pressure to vent. Follow all MSDS/label precautions even after containers are emptied because they may retain product residues.

STORAGE: Keep away from heat, sparks, and flame. Store containers in a cool, well ventilated place. Keep container closed when not in use. Protect from direct sunlight.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Ingredients with Occupational Exposure Limits:

CAS No	Name	ACGIH TLV-TWA	ACGIH-TLV STEL	OSHA PEL-TWA	OSHA PEL-CEILING
78-93-3	2-butanone	200 ppm	300 ppm	200 ppm	N.D.
67-64-1	2-propanone	500 ppm	750 ppm	1000 ppm	N.D.

Personal Protection

	RESPIRATORY PROTECTION: NIOSH/MSHA approved respirators may be necessary if airborne concentrations are expected to exceed exposure limits.
	SKIN PROTECTION: Wear impervious gloves to prevent contact with the skin. Wear protective gear as needed - apron, suit, boots.
	EYE PROTECTION: Wear safety glasses with side shields (or goggles) and a face shield.
	OTHER PROTECTIVE EQUIPMENT: Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.
	HYGIENIC PRACTICES: Do not eat, drink, or smoke in areas where this material is used. Avoid breathing vapors. Remove contaminated clothing and wash before reuse. Wash thoroughly after handling. Wash hands before eating.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance

Liquid

Colour

colorless, clear

Odour

like ketone

Odour threshold

No information available.

Density

~0.809 g/cm³

pH value

N.D.

Viscosity dynamic

N.D.

Solubility in water

Partial

Lower explosion limit

1,8 Vol-%

Upper explosion limit

12,8 Vol-%

Boiling Range

133-178°F (56-81°C)

Flash point

10°F (-14°C)

Evaporation rate

N.D.

Auto ignition temperature

N.D.

10. STABILITY AND REACTIVITY

STABILITY: No Information

CONDITIONS TO AVOID: Avoid impact, friction, heat, sparks, flame and source of ignition.

INCOMPATIBILITY: Prevent contact with halogens. Prevent contact with strong oxidizing agents. Avoid contact with strong reducing agents. Keep away from acids. Keep separate from alkalis.

HAZARDOUS DECOMPOSITION PRODUCTS: During combustion carbon monoxide may be formed. During combustion carbon dioxide may be formed. During combustion asphyxiants may be formed.

HAZARDOUS POLYMERIZATION: No Information

11. TOXICOLOGICAL INFORMATION



Information on Toxicological Effects

Primary Route(s) of Entry: Eye Contact, Ingestion, Inhalation, Skin Contact

EFFECTS OF OVEREXPOSURE - INHALATION: Breathing in the material may irritate the mucous membranes of the nose, throat bronchi and lungs. May cause central nervous system depression. Adverse symptoms may include the following: nausea or vomiting, headache, drowsiness/fatigue, dizziness/vertigo, unconsciousness. Can cause nausea, dizziness, headache, and stupor. Vapor concentrations above recommended exposure levels are irritating to the respiratory tract. Concentrations above recommended exposure levels may cause anesthesia, unconsciousness and other central nervous effects including death.

EFFECTS OF OVEREXPOSURE - SKIN CONTACT: Prolonged or repeated contact can result in defatting and drying of the skin which may result in skin irritation and dermatitis (rash).

EFFECTS OF OVEREXPOSURE - EYE CONTACT: Causes serious irritation to eyes. May cause tearing, redness and discomfort.

EFFECTS OF OVEREXPOSURE - INGESTION: Ingestion may result in nausea, vomiting, diarrhea and pain. May cause central nervous system depression. May cause dizziness and drowsiness and/or stupor.

EFFECTS OF OVEREXPOSURE - CHRONIC HAZARDS: Possible brain damage from overexposure. Small amounts of liquid aspirated into the lungs during ingestion or from vomiting may cause chemical pneumonitis or pulmonary edema. Exposure may damage liver and kidneys. Significant exposure to this chemical may adversely affect people with chronic disease of the respiratory system, central nervous system, kidney, liver, skin, and/or eyes.

Carcinogenicity

CAS-No.	Name	IARC	NTP
78-93-3	2-butanone	3- Not classifiable as to its carcinogenicity to humans.	Not Listed
67-64-1	2-propanone	3- Not classifiable as to its carcinogenicity to humans.	Not Listed

Acute Toxicity Values

The acute effects of this product have not been tested. Data on individual components are tabulated below:

CAS-No.	Name	Oral LD50, mg/kg	Dermal LD50, mg/kg	Vapor LC50, mg/L
78-93-3	2-butanone	>2193	>5000	>20
67-64-1	2-propanone	5800	>2000	50.1

12. ECOLOGICAL INFORMATION

ECOLOGICAL INFORMATION: No Information

13. DISPOSAL CONSIDERATIONS

For more guidance and information contact our Waste Services Division at (262) 658-4000.

Always dispose of any waste in accordance with all local, state, and federal regulations.

DISPOSAL METHOD: Dispose of waste in accordance with all local, state and federal regulations.

STEPS TO BE TAKEN IN CASE MATERIAL IS RELEASED OR SPILLED: Wear appropriate personal protective equipment. (See Exposure Controls / Personal Protection Section.) Eliminate all ignition sources. Evacuate unnecessary personnel. Prevent additional discharge of material if able to do so safely. May cause slipping hazard. Do not touch or walk through spilled material. Avoid runoff into storm sewers and ditches which lead to waterways. Ventilate spill area. Stay upwind of spill. A vapor suppressing foam may be used to reduce vapors. Collect spilled materials for disposal. Use only non-combustible material for clean-up. Use clean, non-sparking tools to collect absorbed materials. Absorb spill with inert material (e.g. dry sand or earth), then place in a chemical waste container. Recover by pumping (use an explosion proof or hand pump).

14. TRANSPORT INFORMATION

DOT proper shipping name:	PRINTING INK RELATED MATERIAL
Packing group:	II
DOT Hazard Class:	3
Hazard Sub Class:	No Information
DOT UN/NA Number:	UN1210
Resp. Guide Page:	129

15. REGULATORY INFORMATION**U.S. Federal Regulations:****CERCLA - SARA Hazard Category**

This product has been reviewed according to the EPA 'Hazard Categories' promulgated under Sections 311 and 312 of the Superfund Amendment and Reauthorization Act of 1986 (SARA Title III) and is considered, under applicable definitions, to meet the following categories:

Fire Hazard, Acute Health Hazard

SARA SECTION 313:

This product contains the following substances subject to the reporting requirements of Section 313 of Title III of the Superfund Amendment and Reauthorization Act of 1986 and 40 CFR part 372:

No SARA 313 components exist in this product.

TOXIC SUBSTANCES CONTROL ACT:

This product contains the following chemical substances subject to the reporting requirements of TSCA 12(B) if exported from the United States:

No TSCA components exist in this product.

U.S. State Regulations:**NEW JERSEY RIGHT-TO-KNOW:**

The following materials are non-hazardous, but are among the top five components in this product.

No NJ Right-To-Know components exist in this product.

PENNSYLVANIA RIGHT-TO-KNOW

The following non-hazardous ingredients are present in the product at greater than 3%.

No PA Right-To-Know components exist in this product.

CALIFORNIA PROPOSITION 65 CARCINOGENS

Warning: The following ingredients present in the product are known to the state of California to cause Cancer.

No Proposition 65 Carcinogens exist in this product.

CALIFORNIA PROPOSITION 65 REPRODUCTIVE TOXINS

Warning: The following ingredients present in the product are known to the state of California to cause birth defects, or other reproductive hazards.

No Proposition 65 Reproductive Toxins exist in this product.

International Regulations: As follows -**CANADIAN WHMIS:**

This SDS has been prepared in compliance with Controlled Product Regulations except for the use of the 16 headings.

WHMIS Class: No Information

16. OTHER INFORMATION**HMIS Ratings:**

Health: 2

Flammability: 3

Reactivity: 0

Personal Protection: X

Volatile Organic Compounds, gr/ltr: 784

DISCLAIMER: THE VOLATILE ORGANIC COMPOUND (VOC) CONTENT REPORTED HEREIN, IF ANY, IS BASED ON A MATERIAL VOC CALCULATION. NOTE THAT SEVERAL METHODS ARE USED FOR CALCULATING VOC CONTENT AND THAT STANDARDS/ REQUIREMENTS REGARDING VOC CONTENT VARY BY LOCATION/JURISDICTION.

ACCORDINGLY, LEIBINGER MAKES NO REPRESENTATIONS OR WARRANTIES, EXPRESS OR IMPLIED, REGARDING THIS MATERIAL'S COMPLIANCE WITH VOC STANDARDS/ REQUIREMENTS APPLICABLE IN LOCATIONS/JURISDICTIONS WHERE THIS MATERIAL MAY BE SOLD OR USED.

Text for GHS Hazard Statements shown in Section 3 describing each ingredient:

H225 Highly flammable liquid and vapor.

H319 Causes serious eye irritation.

H336 May cause drowsiness or dizziness.

Icons for GHS Pictograms shown in Section 3 describing each ingredient:

GHS02



GHS07

Legend: N.A. - Not Applicable, N.E. - Not Established, N.D. - Not Determined, N.I. - No Information

The information on this SDS was obtained from sources which we believe to be reliable. However, the information provided is without any warranty, expressed or implied, regarding its correctness. Some information presented and conclusions drawn herein are from sources other than direct test data on the product itself. The information and recommendations are offered for the user's consideration and examination and should be used to make an independent determination of the methods to safeguard workers and the environment. The conditions or methods of handling, storage, use and disposal of the product are beyond our control and may be beyond our knowledge. For these reasons we do not assume responsibility and expressly disclaim any liability for loss, damage, or expense arising out of or in any way connected with handling, storage, use, or disposal of this product. If the product is used as a component in another product, this SDS may not be applicable. It is the responsibility of the user to comply with all Federal, State, and Local laws and regulations.